

Homework Set 2. Laplace Transform and its Inverse, Transfer Function Model (Solutions)

Question 1

(a)

Let $z_1 = r_1 e^{i\theta_1}$, and $z_2 = r_2 e^{i\theta_2}$.

- $|z_1 z_2| = |z_1| \cdot |z_2|$

LHS:

$$|z_1 z_2| = |r_1 r_2 e^{i(\theta_1 + \theta_2)}| = r_1 r_2 |e^{i(\theta_1 + \theta_2)}| = r_1 r_2$$

RHS:

$$|z_1| \cdot |z_2| = |r_1 e^{i\theta_1}| \cdot |r_2 e^{i\theta_2}| = r_1 |e^{i\theta_1}| \cdot r_2 |e^{i\theta_2}| = r_1 r_2$$

$$LHS = RHS \quad (\text{Proved})$$

- $\angle(z_1 z_2) = \angle z_1 + \angle z_2$

LHS:

$$\angle(z_1 z_2) = \angle(r_1 r_2 e^{i(\theta_1 + \theta_2)}) = \theta_1 + \theta_2$$

RHS:

$$\angle z_1 + \angle z_2 = \angle(r_1 e^{i\theta_1}) + \angle(r_2 e^{i\theta_2}) = \theta_1 + \theta_2$$

$$LHS = RHS \quad (\text{Proved})$$

- $\left| \frac{z_1}{z_2} \right| = |z_1| / |z_2|$ (whenever z_2 is non-zero)

LHS:

$$\left| \frac{z_1}{z_2} \right| = \left| \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} \right| = \frac{r_1}{r_2} |e^{i(\theta_1 - \theta_2)}| = \frac{r_1}{r_2}$$

RHS:

$$|z_1| / |z_2| = |r_1 e^{i\theta_1}| / |r_2 e^{i\theta_2}| = r_1 |e^{i\theta_1}| / r_2 |e^{i\theta_2}| = \frac{r_1}{r_2}$$

$$LHS = RHS \quad (\text{Proved})$$

- $\angle\left(\frac{z_1}{z_2}\right) = \angle z_1 - \angle z_2$

LHS:

$$\angle\left(\frac{z_1}{z_2}\right) = \angle\left(\frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}\right) = \theta_1 - \theta_2$$

RHS:

$$\angle z_1 - \angle z_2 = \angle(r_1 e^{i\theta_1}) - \angle(r_2 e^{i\theta_2}) = \theta_1 - \theta_2$$

$$LHS = RHS \quad (\text{Proved})$$

(b)

Let $z_1 = r_1 e^{i\theta_1}$, and $z_2 = r_2 e^{i\theta_2}$.

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)] \quad (1)$$

We can also write:

$$z_1 = r_1 \cos(\theta_1) + i r_1 \sin(\theta_1) \quad (2)$$

$$z_2 = r_2 \cos(\theta_2) + i r_2 \sin(\theta_2) \quad (3)$$

From (2) and (3), we have:

$$z_1 z_2 = r_1 r_2 [\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 + i(\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2)] \quad (4)$$

Comparing (1), and (4), we have:

$$\cos(\theta_1 + \theta_2) = \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2,$$

$$\sin(\theta_1 + \theta_2) = \cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2.$$

(c)

Let $z_1 = r_1 e^{i\theta_1} = 1 + i$, and $z_2 = r_2 e^{i\theta_2} = \sqrt{3} + 3i$;

$$r_1 = \sqrt{2}; \theta_1 = \tan^{-1}(1) = \frac{\pi}{4} \text{ radians.}$$

$$r_2 = \sqrt{12}; \theta_2 = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \text{ radians.}$$

$$\frac{1+i}{\sqrt{3}+3i} = \frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)} = \frac{1}{\sqrt{6}} e^{-i(\frac{\pi}{12})}$$

Question 2

(a)

With $f(t) = 5 + \exp(-3t) + t \exp(-4t) = 5 S(t) + \exp(-3t) + t \exp(-4t)$ for $t \geq 0$, where $S(t)$ is the unit step function:

$$\begin{aligned} \mathcal{L}[f(t)] &= \mathcal{L}[5 S(t) + \exp(-3t) + t \exp(-4t)] \\ &= 5\mathcal{L}[S(t)] + \mathcal{L}[\exp(-3t)] + \mathcal{L}[t \exp(-4t)] \\ &= 5\mathcal{L}[S(t)] + \mathcal{L}[\exp(-3t)] + \mathcal{L}\left[\frac{t^{2-1}}{(2-1)!} \exp(-4t)\right] \\ &= \boxed{\frac{5}{s} + \frac{1}{s+3} + \frac{1}{(s+4)^2}} \quad (\text{using transforms 2, 5, and 7 from Table 3.1}) \end{aligned}$$

(b)

With $f(t) = (t-1) \cos(4(t-1)) S(t-1) + t^2$, then

$$\begin{aligned} \mathcal{L}[f(t)] &= \mathcal{L}[(t-1) \cos(4(t-1)) S(t-1) + t^2] \\ &= \mathcal{L}[(t-1) \cos(4(t-1))] + \mathcal{L}[t^2] \end{aligned}$$

Let $g(t) = t \cos(4t)$, then $g(t-\theta) = (t-\theta) \cos(4(t-\theta))$

$$\begin{aligned} \mathcal{L}[f(t)] &= \mathcal{L}[g(t-1)S(t-1)] + \mathcal{L}[t^2] \\ &= \mathcal{L}[g(t-1)S(t-1)] + \mathcal{L}[t^{3-1}] \end{aligned}$$

$$\begin{aligned}
&= \mathcal{L}[g(t-1)S(t-1)] + \frac{(3-1)!}{s^3} \text{ (using transforms 4 and 26 from Table 3.1)} \\
&= e^{-s}G(s) + \frac{2}{s^3}
\end{aligned}$$

Where $G(s) = \mathcal{L}[g(t)] = \mathcal{L}[t \cos(4t)] = \int_0^\infty t \cos(4t) e^{-st} dt$. Using an integral table, or integration by parts, we arrive at the following:

$$G(s) = \frac{s^2 - 16}{s^4 + 32s^2 + 256}$$

Thus,

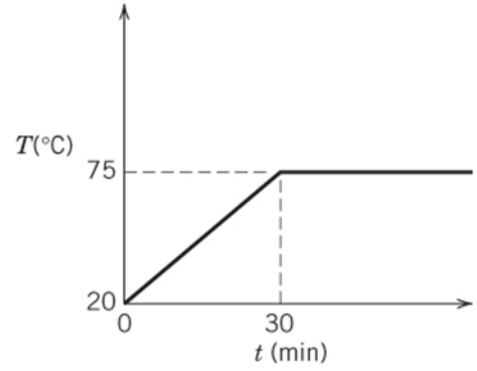
$$\mathcal{L}[f(t)] = \boxed{e^{-s} \left(\frac{s^2 - 16}{s^4 + 32s^2 + 256} \right) + \frac{2}{s^3}}$$

(c)

Using the figure, we can find the define a piecewise function of Temperature with time as follows:

$$\begin{aligned}
T(t) &= \begin{cases} \frac{75-20}{30}t + 20 & 0 \leq t < 30 \\ 75 & t \geq 30 \end{cases} \\
&= \begin{cases} \frac{11}{6}t + 20 & 0 \leq t < 30 \\ 75 & t \geq 30 \end{cases}
\end{aligned}$$

This piecewise function can be expanded into a single-line equation using time-delayed step function:



$$\begin{aligned}
T(t) &= \frac{11}{6}t + 20 + \left\{ -\left(\frac{11}{6}t + 20\right) + 75 \right\} S(t-30) \\
T(t) &= \frac{11}{6}t + 20 + \left\{ -\frac{11}{6}t + 55 \right\} S(t-30) \\
&= \frac{11}{6}t + 20 + \left\{ -\frac{11}{6}(t-30) - \frac{11}{6}(30) + 55 \right\} S(t-30) \\
&= \frac{11}{6}t + 20 - \frac{11}{6}(t-30)S(t-30)
\end{aligned}$$

Since $t \geq 0$,

$$T(t) = \frac{11}{6}t + 20 \text{ } \color{red}{S(t)} - \frac{11}{6}(t-30)S(t-30)$$

Then,

$$\begin{aligned}
\mathcal{L}[T(t)] &= \mathcal{L} \left[\frac{11}{6}t + 20 S(t) - \frac{11}{6}(t-30)S(t-30) \right] \\
&= \frac{11}{6} \mathcal{L}[t] + 20 \mathcal{L}[S(t)] - \frac{11}{6} \mathcal{L}[(t-30)S(t-30)]
\end{aligned}$$

$$= \frac{11}{6} \left(\frac{1}{s^2} \right) + 20 \left(\frac{1}{s} \right) - \frac{11}{6} \left(\frac{e^{-30s}}{s^2} \right) \text{ (using transforms 2, 3, and 26 from Table 3.1)}$$

$$= \boxed{\frac{11}{6} \left(\frac{1 - e^{-30s}}{s^2} \right) + \frac{20}{s}}$$

Question 3

(a)

$$X(s) = \frac{s(s+1)}{(s+2)(s+3)(s+4)} = \frac{\alpha_1}{s+2} + \frac{\alpha_2}{s+3} + \frac{\alpha_3}{s+4}$$

$$\alpha_1 = \left. \frac{s(s+1)}{(s+3)(s+4)} \right|_{s=-2} = 1$$

$$\alpha_2 = \left. \frac{s(s+1)}{(s+2)(s+4)} \right|_{s=-3} = -6$$

$$\alpha_3 = \left. \frac{s(s+1)}{(s+2)(s+3)} \right|_{s=-4} = 6$$

$$X(s) = \frac{1}{s+2} - \frac{6}{s+3} + \frac{6}{s+4} \quad \text{and} \quad x(t) = e^{-2t} - 6e^{-3t} + 6e^{-4t}$$

(b)

$$X(s) = \frac{s+2}{(s+1)^2} = \frac{\alpha_1}{s+1} + \frac{\alpha_2}{(s+1)^2} \quad (1)$$

$$\alpha_2 = (s+2) \Big|_{s=-1} = 1$$

In Eq. 1, substitute any $s \neq -1$ to determine α_1 . Arbitrarily using $s=0$, Eq. 1 gives

$$\frac{2}{1^2} = \frac{\alpha_1}{1} + \frac{1}{1^2} \quad \text{or} \quad \alpha_1 = 1$$

$$X(s) = \frac{1}{s+1} + \frac{1}{(s+1)^2} \quad \text{and} \quad x(t) = e^{-t} + te^{-t}$$

(c)

$$X(s) = \frac{1}{s^2 + s + 1} = \frac{1}{\left(s + \frac{1}{2}\right)^2 + \frac{3}{4}} = \frac{1}{(s+b)^2 + \omega^2}$$

$$\text{where } b = \frac{1}{2} \quad \text{and} \quad \omega = \frac{\sqrt{3}}{2}$$

$$x(t) = \frac{1}{\omega} e^{-bt} \sin \omega t = \frac{2}{\sqrt{3}} e^{-\frac{t}{2}} \sin \frac{\sqrt{3}}{2} t$$

Question 4

The initial conditions: $\frac{d^2x}{dt^2}(0) = 0$, $\frac{dx}{dt}(0) = 0$, $x(0) = 0$

(a)

$$\frac{d^3x}{dt^3} + 2\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x = 3$$

Do Laplace transform:

$$\begin{aligned} & \left[s^3 \cdot X(s) - s^2 \cdot x(0) - s \cdot \frac{dx}{dt}(0) - \frac{d^2x}{dt^2}(0) \right] + 2 \left[s^2 \cdot X(s) - s \cdot x(0) - \frac{dx}{dt}(0) \right] \\ & + 2[s \cdot X(s) - x(0)] + X(s) = \frac{3}{s} \\ & s^3 \cdot X(s) + 2s^2 \cdot X(s) + 2s \cdot X(s) + X(s) = \frac{3}{s} \\ X(s) &= \frac{3}{s(s^3 + 2s^2 + 2s + 1)} = \frac{3}{s(s+1)(s + \frac{1}{2} + \frac{\sqrt{3}}{2}i)(s + \frac{1}{2} - \frac{\sqrt{3}}{2}i)} \\ X(s) &= \frac{3}{s(s+1)(s + \frac{1}{2} + \frac{\sqrt{3}}{2}i)(s + \frac{1}{2} - \frac{\sqrt{3}}{2}i)} = \frac{A}{s} + \{\text{other terms}\} \end{aligned}$$

The pole at $s=0$ indicates a constant term in the time domain suggesting that the system has a steady-state component that does not decay over time. The other poles are on the LHP as they all have negative real parts, indicating exponential decay to zero in the time domain. Hence $x(t)$ converges (to a non-zero constant).

(b)

$$\frac{d^2x}{dt^2} - x = 2e^t$$

The Laplace transform:

$$\left[s^2 \cdot X(s) - s \cdot x(0) - \frac{dx}{dt}(0) \right] - X(s) = \frac{2}{s-1}$$

$$s^2 \cdot X(s) - X(s) = \frac{2}{s-1}$$

$$X(s) = \frac{2}{(s-1)(s^2-1)} = \frac{2}{(s-1)^2(s+1)}$$

The denominator does not have complex roots, so function $x(t)$ is not oscillatory.
The system has a pole at $s=1$ in the right half-plane. Hence, $x(t)$ diverges.

(c)

$$\frac{d^3x}{dt^3} + x = \sin(t)$$

The Laplace transform:

$$\left[s^3 \cdot X(s) - s^2 \cdot x(0) - s \cdot \frac{dx}{dt}(0) - \frac{d^2x}{dt^2}(0) \right] + X(s) = \frac{1}{s^2+1}$$

$$s^3 \cdot X(s) + X(s) = \frac{1}{s^2+1}$$

$$X(s) = \frac{1}{(s^2+1)(s^3+1)} = \frac{1}{(s+i)(s-i)(s+1)(s-\frac{1}{2}+\frac{\sqrt{3}}{2}i)(s-\frac{1}{2}-\frac{\sqrt{3}}{2}i)}$$

The denominator has complex roots, so $x(t)$ is oscillatory.

The system has poles at $s = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$ (in the right half-plane). Hence, $x(t)$ diverges.

(d)

$$\frac{d^2x}{dt^2} + x = 4$$

The Laplace transform:

$$\left[s^2 \cdot X(s) - s \cdot x(0) - \frac{dx}{dt}(0) \right] + X(s) = \frac{4}{s}$$

$$s^2 \cdot X(s) + X(s) = \frac{4}{s}$$

$$X(s) = \frac{4}{s(s^2+1)} = \frac{4}{s(s+i)(s-i)}$$

The denominator has complex roots, leading to persistent oscillations in $x(t)$. The pole at $s=0$ introduces a constant term in the time domain, while the purely imaginary poles cause undamped oscillations. Hence, $x(t)$ does not converge and remains oscillatory.

[\[x\(t\)=4-4cos\(t\)\]](#)

Question 5

The transfer function for the pressure transmitter is given by,

$$\frac{P_m(s)}{P(s)} = \frac{1}{(30s+1)}$$

And $P'(s) = 15/s$ for the step change from 35 to 50 psi.

$$\frac{P'_m(s)}{P'(s)} = \frac{1}{(30s+1)} \cdot \frac{15}{s}$$

The step response after taking the Laplace inverse is given by-

$$P'_m(t) = 15(1 - e^{-\frac{t}{30}})$$

Let t_a be the time that the alarm sounds. Then,

$$P'_m(t_a) = 45 - 35 = 10 \text{ psi.}$$

$$10 = 15(1 - e^{-\frac{t}{30}})$$

$$t_a = 32.98 \text{ s}$$

The alarm will sound 32.98 s after 1:30 PM.

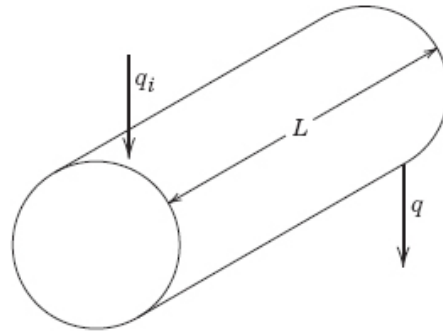
Question 6

Setting Up Balance Equation with Mass Balance

$$\{\text{accumulation}\} = \{\text{in}\} - \{\text{out}\}$$

$$\frac{dm}{dt} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}}$$

$$\frac{d(\rho V)}{dt} = \rho_i q_i - \rho q$$



Assuming constant density, $\rho(t) = \rho = \rho_i$

$$\frac{dV}{dt} = q_i - q$$

Since the volume of the water accumulated in the tank at a given height is given by: $V =$

$$\left[R^2 \arccos\left(1 - \frac{h}{R}\right) - (R - h)\sqrt{R^2 - (R - h)^2} \right] L, \text{ then,}$$

$$\frac{d\left(\left[R^2 \arccos\left(1 - \frac{h}{R}\right) - (R - h)\sqrt{R^2 - (R - h)^2}\right] L\right)}{dt} = q_i - q$$

$$L \left(\frac{d\left(R^2 \arccos\left(1 - \frac{h}{R}\right)\right)}{dt} - \frac{d\left((R - h)\sqrt{R^2 - (R - h)^2}\right)}{dt} \right) = q_i - q$$

$$R^2 \frac{d\left(\arccos\left(1 - \frac{h}{R}\right)\right)}{dt} - (R - h) \frac{d\left(\sqrt{R^2 - (R - h)^2}\right)}{dt} - \sqrt{R^2 - (R - h)^2} \frac{d(R - h)}{dt} = \frac{q_i - q}{L}$$

Let $u = 1 - \frac{h}{R}$ and $v = R^2 - (R - h)^2$ where $\frac{du}{dt} = \frac{-1}{R} \frac{dh}{dt}$ and $\frac{dv}{dt} = 2(R - h) \frac{dh}{dt}$, then

$$\begin{aligned}
R^2 \frac{d(\arccos(u))}{dt} \frac{du}{dt} - (R-h) \frac{d(\sqrt{v})}{dt} \frac{dv}{dt} + \sqrt{R^2 - (R-h)^2} \frac{dh}{dt} &= \frac{q_i - q}{L} \\
R^2 \left(\frac{-1}{\sqrt{1-u^2}} \right) \left(\frac{-1}{R} \frac{dh}{dt} \right) - (R-h) \left(\frac{1}{2\sqrt{v}} \right) \left(2(R-h) \frac{dh}{dt} \right) + \sqrt{R^2 - (R-h)^2} \frac{dh}{dt} &= \frac{q_i - q}{L} \\
R \left(\frac{1}{\sqrt{1 - \left(1 - \frac{h}{R}\right)^2}} \right) \left(\frac{dh}{dt} \right) - \left(\frac{(R-h)^2}{\sqrt{R^2 - (R-h)^2}} \right) \left(\frac{dh}{dt} \right) + \sqrt{R^2 - (R-h)^2} \left(\frac{dh}{dt} \right) &= \frac{q_i - q}{L} \\
\left(\frac{R^2}{\sqrt{R^2 - (R-h)^2}} \right) \left(\frac{dh}{dt} \right) - \left(\frac{(R-h)^2}{\sqrt{R^2 - (R-h)^2}} \right) \left(\frac{dh}{dt} \right) + \sqrt{R^2 - (R-h)^2} \left(\frac{dh}{dt} \right) &= \frac{q_i - q}{L} \\
2 \left(\frac{R^2 - (R-h)^2}{\sqrt{R^2 - (R-h)^2}} \right) \left(\frac{dh}{dt} \right) &= \frac{q_i - q}{L} \\
\sqrt{R^2 - (R-h)^2} \left(\frac{dh}{dt} \right) &= \frac{q_i - q}{L} \\
\boxed{\frac{dh}{dt} = \left(\frac{q_i(t) - q(t)}{2L} \right) \left(\frac{1}{\sqrt{R^2 - (R-h(t))^2}} \right)}
\end{aligned}$$

Find Steady-State Condition

Let $\bar{h}$, $\bar{q}_i$, and $\bar{q}$ represent the steady state conditions of height, inlet flowrate, and exit flowrate, respectively. Steady state is found by setting the time derivative to zero:

$$0 = \left(\frac{\bar{q}_i - \bar{q}}{2L} \right) \left(\frac{1}{\sqrt{R^2 - (R - \bar{h})^2}} \right)$$

Or

$$\bar{q}_i = \bar{q}$$

Tank is at steady state when the inlet and outlet flowrates are equal. Notice the steady-state condition is independent of height; thus, this steady-state condition is applicable to any height. Since we are stating that the tank is initially at steady-state condition, the steady-state height is its initial height.

Model in Deviation Variable Form

Let $h' = h - \bar{h}$, $q'_i = q_i - \bar{q}_i$, and $q' = q - \bar{q}$, then

$$\frac{d(h' + \bar{h})}{dt} = \left(\frac{q'_i + \bar{q}_i - q' - \bar{q}}{2L} \right) \left(\frac{1}{\sqrt{R^2 - (R - h' - \bar{h})^2}} \right)$$

Since $\bar{q}_i = \bar{q}$

$$\frac{d(h')}{dt} = \left(\frac{q'_i - q'}{2L} \right) \left(\frac{1}{\sqrt{R^2 - (R - h' - \bar{h})^2}} \right)$$

Linearizing the Model using Taylor Series Expansion

The model is in the form $\frac{dh'}{dt} = f(h', q'_i, q')$, and it can be linearized as $\frac{dh'}{dt} \approx \left. \frac{\partial f}{\partial h'} \right|_s h' + \left. \frac{\partial f}{\partial q'_i} \right|_s q'_i +$

$\left. \frac{\partial f}{\partial q'} \right|_s q'$ where s denotes that the partial derivative is evaluated at the steady-state condition:

($h' = 0, q'_i = 0, q' = 0$)

$$\begin{aligned}
\frac{d(h')}{dt} &\approx \left(\frac{q'_i - q'}{2L} \right) \frac{\partial}{\partial h'} \left\{ \frac{1}{\sqrt{R^2 - (R - h' - \bar{h})^2}} \right\} \bigg|_s h' + \left(\frac{1}{2L\sqrt{R^2 - (R - h' - \bar{h})^2}} \right) \frac{\partial}{\partial q'_i} \{q'_i - q'\} \bigg|_s q'_i \\
&\quad + \left(\frac{1}{2L\sqrt{R^2 - (R - h' - \bar{h})^2}} \right) \frac{\partial}{\partial q'} \{q'_i - q'\} \bigg|_s q' \\
\frac{d(h')}{dt} &\approx \left(\frac{0 - 0}{2L} \right) \frac{\partial}{\partial h'} \left\{ \frac{1}{\sqrt{R^2 - (R - h' - \bar{h})^2}} \right\} \bigg|_s h' + \left(\frac{1}{2L\sqrt{R^2 - (R - 0 - \bar{h})^2}} \right) \frac{\partial}{\partial q'_i} \{q'_i - q'\} \bigg|_s q'_i \\
&\quad + \left(\frac{1}{2L\sqrt{R^2 - (R - 0 - \bar{h})^2}} \right) \frac{\partial}{\partial q'} \{q'_i - q'\} \bigg|_s q'
\end{aligned}$$

The first term on the right-hand side goes to zero.

$$\boxed{\frac{d(h')}{dt} \approx \left(\frac{1}{2L\sqrt{R^2 - (R - \bar{h})^2}} \right) q'_i - \left(\frac{1}{2L\sqrt{R^2 - (R - \bar{h})^2}} \right) q'}$$

Question 7

1. Ball Valves:

- Mechanism: Operates with a spherical ball that has a bore through its center. The valve opens when this bore aligns with the flow and closes with a 90-degree rotation, effectively obstructing the flow.
- Applications: Favored for its reliable sealing in high pressure and temperature settings, typically used in water, natural gas, and oil processing for secure and tight shutoff.

2. Butterfly Valves:

- Mechanism: Features a thin disc mounted on a rod that rotates to either block or open the flow path.
- Applications: Ideal for HVAC systems and large volume water services due to their quick operation and ability to effectively throttle flow.

3. Plug Valves:

- Mechanism: Uses a cone-shaped or cylindrical plug that rotates within the valve body, with passageways that line up with inlets and outlets to regulate flow.

- Applications: Common in settings requiring frequent operation, such as slurry handling, where minimal flow resistance and prevention of clogging are critical.

4. Rotary Globe Valves:

- Mechanism: Utilizes a rotating disk or rotor inside the valve body to finely control fluid passage.
- Applications: Suitable for precise flow regulation in dosing, mixing, and filling operations, especially in the pharmaceutical and chemical industries where manageable pressure drops are necessary.

Question 8

(a)

$$\frac{\partial x}{\partial t} = \frac{\partial^2 x}{\partial z^2} - \beta x$$

Applying the Laplace transform with respect to time t , we have:

$$sX(z, s) - x(z, 0) = \frac{d^2 X}{dz^2}(z, s) - \beta X(z, s).$$

Assuming the initial condition, $x(z, 0) = 0$, we obtain:

$$\frac{d^2 X}{dz^2}(z, s) = (s + \beta)X(z, s) \quad (1)$$

Converting the boundary conditions on $X(z, s)$:

- $x(z = 0, t) = u(t)$ becomes:

$$X(z = 0, s) = U(s). \quad (2)$$

- $x(z = 1, t) + \frac{\partial x}{\partial z}(z = 1, t) = 0$ becomes:

$$X(1, s) + \frac{dX}{dz}(1, s) = 0 \quad (3)$$

(b)

The solution to (1) is given by:

$$X(z, s) = A \cosh(\sqrt{s + \beta} z) + B \sinh(\sqrt{s + \beta} z), \quad (4)$$

where A and B are to be determined.

Using the boundary condition given in (2),

$$X(0, s) = A \cosh(0) + B \sinh(0) = A = U(s) \quad (5)$$

Using the boundary condition given in (3),

$$A \cosh(\sqrt{s + \beta}) + B \sinh(\sqrt{s + \beta}) + A \sqrt{s + \beta} \sinh(\sqrt{s + \beta}) + B \sqrt{s + \beta} \cosh(\sqrt{s + \beta}) = 0. \quad (6)$$

From (5) and (6), we have:

$$B = -U(s) \frac{\cosh(\sqrt{s + \beta}) + \sqrt{s + \beta} \sinh(\sqrt{s + \beta})}{\sinh(\sqrt{s + \beta}) + \sqrt{s + \beta} \cosh(\sqrt{s + \beta})} \quad (7)$$

With A and B obtained in (5) and (7), we now have the full expression for the solution in (4).

- Obtaining the transfer function:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{X(1, s)}{U(s)} \quad (8)$$

Obtain $X(1, s)$ using (4)

$$X(1, s) = A \cosh(\sqrt{s + \beta}) + B \sinh(\sqrt{s + \beta})$$

Using (5) and (6), we have:

$$X(1, s) = U(s) \left[\cosh(\sqrt{s + \beta}) - \frac{\cosh(\sqrt{s + \beta}) + \sqrt{s + \beta} \sinh(\sqrt{s + \beta})}{\sinh(\sqrt{s + \beta}) + \sqrt{s + \beta} \cosh(\sqrt{s + \beta})} \sinh(\sqrt{s + \beta}) \right] \quad (9)$$

Simplifying (9), while noting that $\cosh^2(\sqrt{s + \beta}) - \sinh^2(\sqrt{s + \beta}) = 1$, we have:

$$X(1, s) = U(s) \left[\frac{\sqrt{s + \beta}}{\sinh(\sqrt{s + \beta}) + \sqrt{s + \beta} \cosh(\sqrt{s + \beta})} \right] \quad (10)$$

Plugging (10) into (8), we finally obtain:

$$G(s) = \frac{\sqrt{s + \beta}}{\sinh(\sqrt{s + \beta}) + \sqrt{s + \beta} \cosh(\sqrt{s + \beta})}.$$